Question 1 - Parts Order Form

Assumptions
- customerNumber identifies a customer; partNumber identifies a part. One order per customer, date and time.
- Each part appears once per order and has one current name, type, cage and unit price. Any employee may help any customer.
- Date/Time, Employee, Name and Type are renamed orderDate/orderTime, employeeName, partName and partType.

Original form field	Value
Customer Number	H G54587
Customer Name	Jeff Peterson
Customer Type	Consumer
Date / Time	7/1/2024 / 10:30am
Employee	D. Harrison
Part line 1	10654 / Float Control / Plumbing / G413 / 4 / 12
Part line 2	10456 / Modulator / Electrical / H433 / 3 / 7
Part line 3	10776 / Hose Assembly / Plumbing / G413 / 7 / 9
Part line 4	10657 / Float Assembly / Plumbing / G413 / 5 / 10

order_record
customerNumber (PK)
orderDate (PK)
orderTime (PK)
partNumber (PK)
customerName
customerType
employeeName
partName
partType
cageCode
quantityOrdered
unitPrice

1NF: flatten the repeating parts
One row represents one part in one order.
Key = customerNumber + orderDate + orderTime + partNumber.
Dependencies
customerNumber → customerName, customerType
partNumber → partName, partType, cageCode, unitPrice
customerNumber + orderDate + orderTime → employeeName
Full key → quantityOrdered

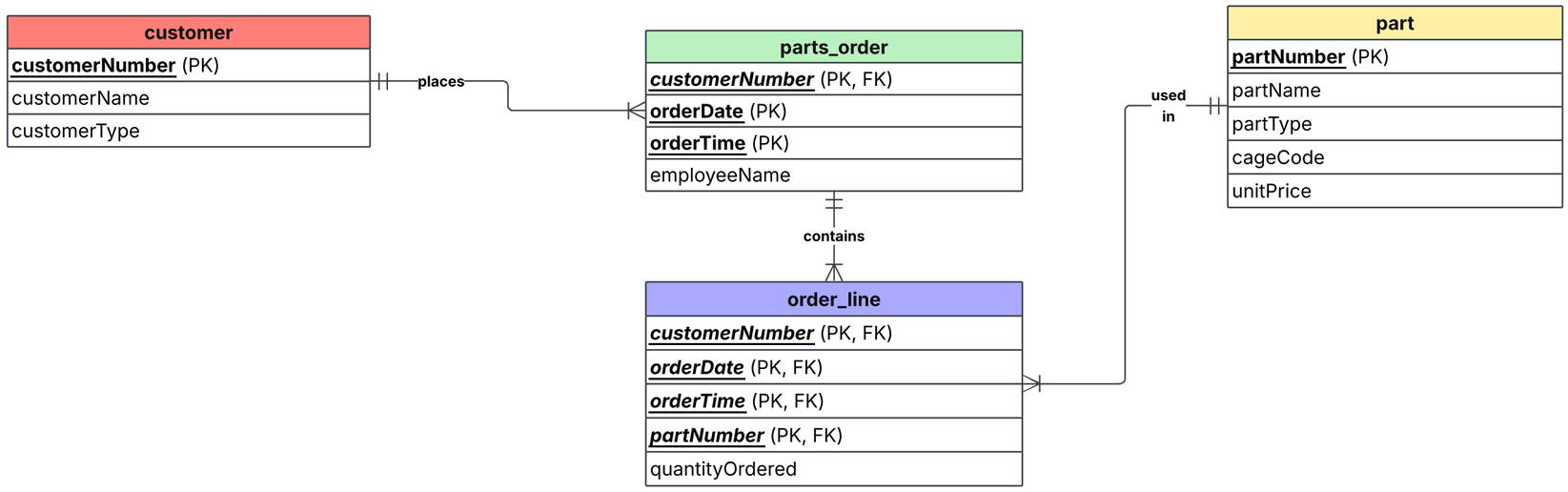
customer
customerNumber (PK)
customerName
customerType

parts_order
customerNumber (PK)
orderDate (PK)
orderTime (PK)
employeeName

part
partNumber (PK)
partName
partType
cageCode
unitPrice

order_line
customerNumber (PK)
orderDate (PK)
orderTime (PK)
partNumber (PK)
quantityOrdered

2NF: remove partial dependencies
Customer facts depend only on customerNumber.
Part facts depend only on partNumber.
employeeName depends on the order key.
quantityOrdered depends on the full line key.
Each non-key value now depends on its whole key.



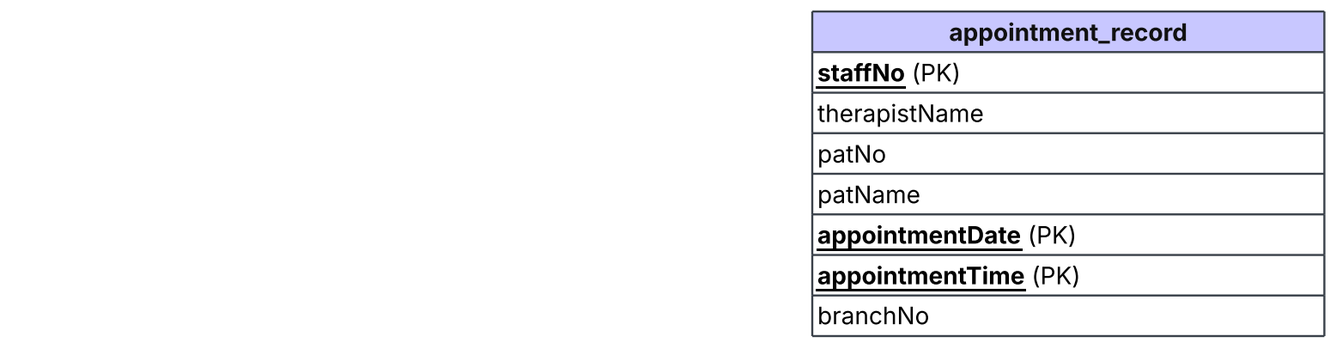
3NF: check transitive dependencies
No remaining non-key attribute determines another non-key attribute under these assumptions.
The 2NF relations are already in 3NF.

Foreign keys
parts_order.customerNumber → customer
order_line.(customerNumber, orderDate, orderTime) → parts_order
order_line.partNumber → part

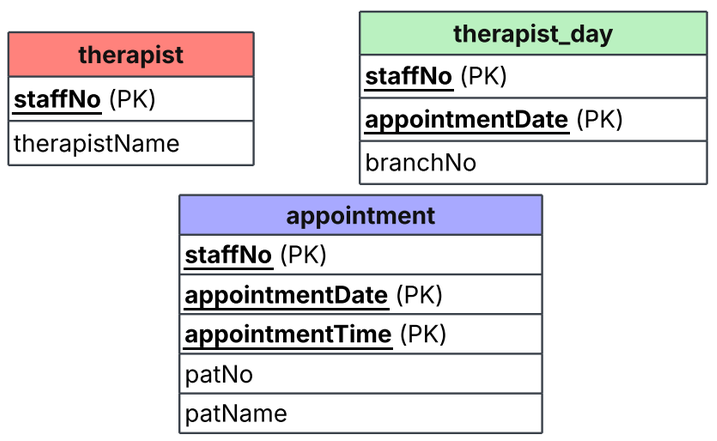
Question 2 - Therapist Appointments

Assumptions
- staffNo identifies a therapist; patNo identifies a patient.
- A therapist has at most one appointment at a given date and time.
- Each appointment has one patient. A therapist works at one branch per day, but may work at other branches on different days. branchNo has no other attributes in the given data.

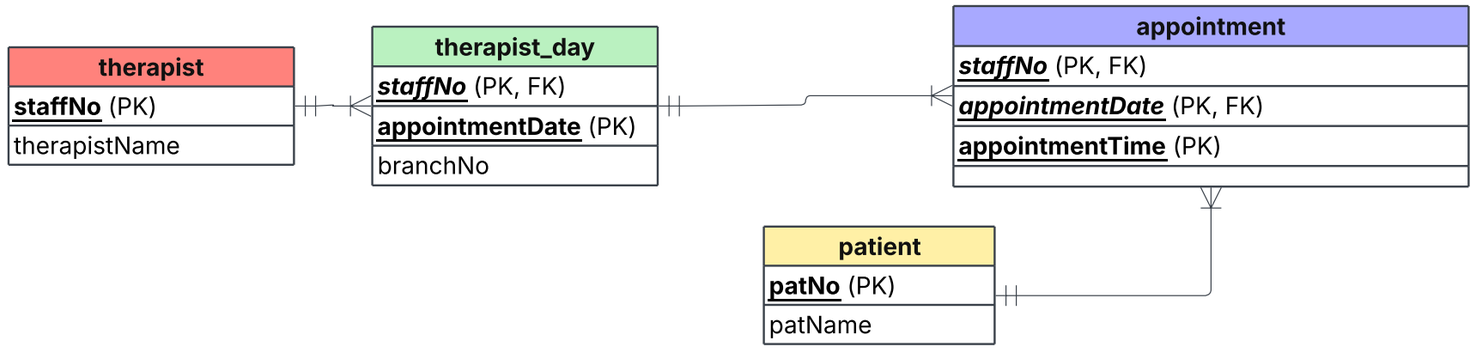
staffNo	therapistName	patNo	patName	appointmentDate	appointmentTime	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13



1NF: atomic appointment rows
The given rows already contain one value per cell.
PK = staffNo + appointmentDate + appointmentTime.
Dependencies
staffNo → therapistName
staffNo + appointmentDate → branchNo
patNo → patName
Full appointment key → patNo



2NF: remove partial dependencies
therapistName depends on staffNo alone.
branchNo depends on staffNo plus date.
Move these facts to therapist and therapist_day.
patName still depends on patNo, so a transitive dependency remains in appointment.



3NF: remove the transitive dependency
patNo → patName, so move patName to patient.
Keep only patNo in appointment.

Foreign keys
therapist_day.staffNo → therapist
appointment.(staffNo, appointmentDate) → therapist_day
appointment.patNo → patient

Question 3 - Event Staff Hours

Assumptions
- Each eNo identifies one employee. Each contractNo belongs to one event.
- Each eventNo identifies one event and its location. Several contracts may refer to the same event.
- Each employee-contract pair has one total hours value.

eNo	contractNo	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

staff_hours
eNo (PK)
contractNo (PK)
hours
eName
eventNo
eventLoc

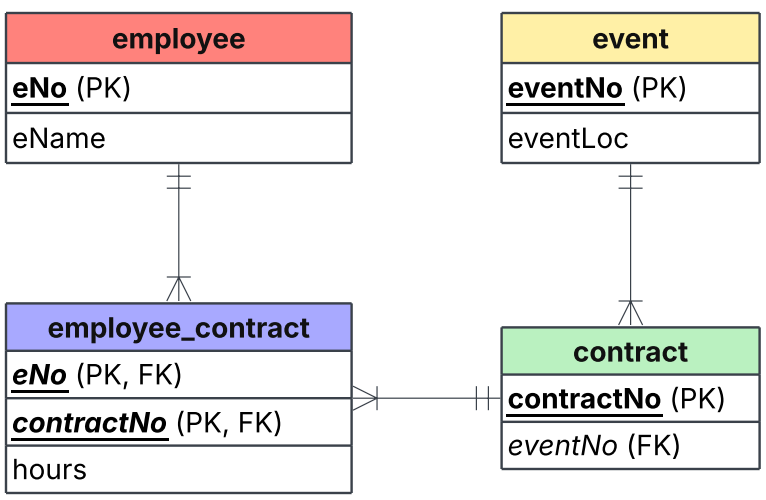
1NF: one row per employee-contract pair
The given rows are already atomic.
PK = eNo + contractNo.
Dependencies
eNo → eName
contractNo → eventNo
eventNo → eventLoc
Full key → hours

employee
eNo (PK)
eName

contract
contractNo (PK)
eventNo
eventLoc

employee_contract
eNo (PK)
contractNo (PK)
hours

2NF: remove partial dependencies
eName depends on eNo alone.
eventNo and eventLoc depend on contractNo.
hours depends on both key fields.
Move employee and contract facts out of the employee_contract relation.
eventNo → eventLoc still remains in contract.



3NF: remove the transitive dependency
contractNo → eventNo → eventLoc.
Move eventLoc to event, keyed by eventNo.
Keep eventNo in contract as an FK.

Foreign keys
contract.eventNo → event
employee_contract.eNo → employee
employee_contract.contractNo → contract